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SUPERSEDING
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May 26, 1959

FEDERAL SPECIFICATION

CLOTH, COTTON, SEERSUCKER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for cloth, cotton seersucker.

1.2 Classification.

1.2.1 *Types and classes.* The cloth, cotton, seersucker covered by this specification shall be of the following types and classes (see 6.2):

Type I.—Narrow stripe.
  Class 1.—82 by 86 texture.
Type II.—Medium stripe.
  Class 1.—112 by 80 texture.
  Class 2.—80 by 80 texture.
Type III.—Stripe pattern effect.
  Class 1.—280 [1] by 72 texture.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and Standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:
CCC–T–191—Textile Test Methods.
PPP–P–51—Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks, and Synthetics).

Federal Standards:
Fed. Std. No. 4—Glossary of Fabric Imperfections.
Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbook as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington 25, D.C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standard:
MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the

[1] Per repeat (approximately 2⅝ inches).

FSC 8305

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extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Rules and Regulations Under The Textile Fiber Products Identification Act effective March 3, 1960.

(Copies may be obtained, free of charge, from the Federal Trade Commission, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the cloth shall match the standard for shade and shall be equal to or better than the standard with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 *Yarn.* The yarn shall be made from cotton which has been thoroughly cleaned, carded, combed, drawn, and evenly spun into single yarns except that the slack beam warp yarn for the type I, class 1 cloth and tight beam warp yarn for the type II, class 2 cloth shall be two-ply.

3.3 Color. The color shall be as specified and, where applicable (see 3.1), shall match an approved standard sample (see 6.2). The use of sulfur dyes or dyes containing sulfur compounds capable of oxidation to form sulfuric acid are prohibited (see 4.4.1).

3.3.1 *Matching.* The color shall match the approved standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard shade under incandescent lamplight at 2800° Kelvin.

3.3.2 *Colorfastness.*

3.3.2.1 *Rating per standard sample.* The finished cloth shall show colorfastness to laundering, bleaching, perspiration, crocking, and light equal to or better than the standard sample when tested as specified in 4.4.

3.3.2.2 *Rating per no standard sample.* The finished cloth shall show good colorfastness to laundering, bleaching, perspiration, crocking, and light when tested as specified in 4.4.

3.4 Physical requirements. The finished cloth shall conform to the requirements shown in table I, when tested as specified in 4.4.

3.4.1 *Weave.*

3.4.1.1 *Types I and III.* The weave shall be plain.

3.4.1.2 *Type II.* The weave shall be plain with the slack warp weaving two ends as one.

3.4.2 *Slack beam take-up.* The minimum slack beam take-up for all types and classes shall be 40 percent, when tested as specified in 4.4.

TABLE I. *Physical requirements*

Type	Class	Weight (ounces per square yard) minimum	Yarns per inch minimum		Breaking strength (grab method) minimum	
			Warp	*Filling*	*Warp*	*Filling*
I	1	4.0	82	86	24	45
II	1	4.0	112	80	30	40
II	2	4.0	80	80	30	40
III	1	4.0	280[1]	72	25	30

[1]Per repeat (see 3.4.3.3).

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3.4.3 *Stripes or pattern effect.*

3.4.3.1 *Type I, class 1.* The stripe effect of the type I cloth shall be produced by using two warp beams, one tight and one slack. The color of the tight beam warp yarn shall be as specified (see 6.2). The slack beam warp yarns shall be natural or bleached. The two warps shall alternate in sections of tight and slack ends and each color pattern repeat shall be as follows:

Tight – 6
Slack – 4

3.4.3.2 *Type II, classes 1 and 2.* The stripe effect of the type II cloth shall be produced by using two warp beams, one tight and one slack. The color of the tight beam warp yarn shall be as specified (see 6.2). The slack beam warp yarn shall be natural or bleached. The two warps shall alternate in sections of tight and slack ends and each color pattern repeat shall be as follows:

Tight – 10
Slack – 12

The reeding plan shall produce a concentration of slack beam warp ends double that of the tight beam ends.

3.4.3.3 *Type III, class 1.* The pattern effect of the type III, class I cloth shall be produced by using two warp beams, one tight and one slack. The color of the slack beam warp yarn shall be as specified (see 6.2). The tight beam warp yarn shall be natural or bleached. The two warps shall alternate in sections of tight and slack ends and each color pattern repeat shall be as follows:

Tight beam–	16	16	16	16	16	8	8	8	8
Slack beam–	12	12	12	12	24	24	24	24	24

The warp reeding plans shall produce finished stripes of the following widths:

Colored yarn 24 ends — $\frac{3}{16} \pm \frac{1}{32}$ inch
Colored yarn 12 ends — $\frac{3}{32} \pm \frac{1}{32}$ inch
Bleached yarn 8 ends — $\frac{3}{32} \pm \frac{1}{32}$ inch
Bleached yarn 16 ends — $\frac{3}{16} \pm \frac{1}{32}$ inch

The total width of each 280 end repeat shall be $2\frac{5}{8} \pm \frac{1}{8}$ inch. The repeats of the patterns shall be balanced between the selvages.

3.5 Width. The width shall be as specified and shall be the minimum width exclusive of selvage (see 6.2).

3.6 Finish. The cloth shall be closely singed, desized, and bleached. When bleached warp and filling yarns are used for the undyed yarns, yarn or piece bleaching is optional. Residual size, finishing, or other nonfibrous material shall not exceed 2 percent when tested as specified in 4.4.

3.7 Shrinkage. The cloth shall be preshrunk and shall not shrink more than 4 percent in the direction of the warp nor more than 2 percent in the direction of the filling when tested as specfied in 4.4. The preshrinking process shall not be identified on the cloth, the ticket or on the package.

3.8 Length. Unless otherwise specified, the cloth shall be in one continuous piece of not less than 50 yards (see 6.2) and shall be put up in folded pieces as specified in PPP–P–51.

3.9 Fiber identification. The material shall be labeled, ticketed or invoiced for fiber content in accordance with the Rules and Regulations Under The Textile Fiber Products Identification Act effective March 3, 1960.

3.10 Workmanship. The cloth shall conform to the quality and grade of product established by this specification, and the occurrence of defects shall not exceed the levels set by the applicable acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Contractor's inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Govern-

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ment as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are

Yard by Yard Examination

Glossary No. of Fed. Std. No. 4	Defects	Classification	
		Major	Minor
79	Abrasion mark:		
	Resulting in a thin or weak place	X	
	Clearly visible [1] but not resulting in a weak place		X
11, 33	Broken or missing yarns:		
	Three or more contiguous regardless of length, two contiguous 2 inches or more in length	X	
	Two contiguous less than 2 inches in length, or single more than 4 inches missing		X
1, 2	Coarse or heavy filling bar—clearly visible [1]	X	
55, 56, 90, 91, 92	Cut, hole or tear (including selvage) Any	X	
9	Fine or light filling bar, light place clearly visible [1]	X	
13, 34	Floats or skips:		
	Multiple, ½ inch or more in combined warp and filling directions	X	
	Multiple, less than ½ inch in combined directions, or single 2 inches or more in length		X
35	Hitchback:		
	Resulting in thin area or weak spot ½ inch or more in combined directions	X	
	Clearly visible [1] but not resulting in a weak spot		X
3, 5	Jerked-in or sloughed-off filling:		
	Two or more extra yarns in the shed:		
	Extending more than 4 inches	X	
	Extending 4 inches or less		X
10	Mispick:		
	Resulting in clearly visible double pick or harness skips		X
14	Mixed filling or shade bar—		
	Clearly visible [1]	X	
41	Reed mark:		
	Resulting in clearly visible [1] warp-wise separation 9 inches or longer, and distinctly wider than normal		X
37	Tight end:		
	Occurring in the slack warp and clearly visible [1] or any slack section having noticeably less take-up than other unaffected slack sections		X
5, 27, 77	Slub, slug or gout:		
	More than three times the thickness of normal yarn, 1 inch or more in length	X	
	Less than 1 inch in length and clearly visible [1]		X
78	Smash—Any	X	
74, 75, 76 [2]	Spot, stain or streak:		
	Clearly visible [1] extending 1 inch or more in combined directions	X	
	Clearly visible [1] but extending less than 1 inch in combined directions		X
	Width—less than specified		X

[1] Clearly visible at normal inspection distance (approximately 3 feet).

[2] Except defects of spots, stain and streaks shall be counted regardless of cause, i.e., not restricted to rust, oil or grease.

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deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Examination of the end item. Except when otherwise specified by Civil Government agencies, inspection shall be in accordance with the provisions of MIL—STD–105 except where otherwise indicated. Defects found during this examination shall be classified in accordance with 4.2.1, 4.2.1.1, and 4.2.1.2.

4.2.1 *Yard-by-yard examination.* The required yardage of each piece shall be examined on the face side and visual defects classified as listed. The defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition of the cloth, in which case only the more serious defects shall be counted. A continuous defect shall be counted as one defect for each warp-wise yard or fraction thereof in which it occurs. The unit of product for this examination shall be 1 linear yard. The acceptable quality level (AQL) for this examination shall be 2.5 major defects and 6.5 total defects (major and minor combined) per hundred units (yards). The sample size shall be based upon inspection level II of MIL–STD–105. The sample size shall be expressed in units of 1 yard each. The number of pieces from which the sample is to be selected shall be in accordance with table II of 4.2.1.1. An approximately equal number of yards shall be examined in each piece of the sample. No more than one piece shall be selected from each container.

4.2.1.1 *Overall examination.* Each defect listed shall be counted not more than once in each piece examined. The unit of product for this examination shall be one piece. The sample size (number of pieces selected as sample) for this examination and the acceptance number (number of defects acceptable) shall be as shown in table II.

Defects

Baggy, ridgy or wavy cloth—clearly noticeable.
Preshrinking process identified by name or trademark on cloth or ticket.
Overall uncleanness.
Incorrect pattern, i.e., color arrangement or width of stripes not as specified.
Pattern not centered against both selvages as specified.
Off shade, not within established tolerances.
Shaded end to end, side to side, side to center, or throughout piece.
Cloudy, streaky or mottled throughout piece.
Uneven weaving throughout.
Selvage beaded, rolled, folded or slack, continuously or intermittently for more than half the length of the piece.

TABLE II. *Sample size*

Lot sizes in yards	Sample size in pieces	Acceptance number
Up to 1,300 [1]	3	0
1,301 to 3,200	5	0
3,201 to 8,000	7	0
8,001 to 22,000	10	0
22,001 to 110,000	15	0
110,001 to 550,000	25	1
550,001 and up	50	2

[1] If a lot contains fewer than three pieces, each piece in the lot shall be examined.

4.2.1.2 *Examination for length.*

4.2.1.2.1 *Individual pieces.* The piece shall be examined for gross length. Any gross length found to be less than the minimum specified or more than 2 yards below the gross length marked on the ticket shall be considered a defect with respect to length. The unit of product for this examination shall be one piece. The number of pieces selected as sample (sample size) for this examination and the acceptance number shall be in accordance with table II.

4.2.1.2.2 *Total yardage in sample.* The lot shall be unacceptable if the total of the actual gross length of pieces in the sample is less than the total of the gross lengths marked on the tickets.

4.3 Examination of preparation for delivery requirements. An examination shall be

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Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
	Preshrinkage process is identified by name or trademark.
Materials	Any component missing.
	Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as: incomplete closure of case liners, container flaps, loose strapping, inadequate stapling.
	Bulging or distortion of containers.
Weight	Gross weight exceeds requirements.

made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list above. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of containers in the inspection lot. The inspection level shall be L–4 and the AQL shall be 4.0 defects per hundred units.

4.4 Testing of the end item. The method of testing specified in CCC–T–191, wherever applicable and as listed in table III, shall be followed. When the data in the "number of determinations per sample unit" and the "results reported as" columns are not specified in the table, they shall be as required by the referenced test method. The physical and chemical values specified in section 3 apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test method. The unit of product shall be 3 continuous yards full width of the finished cloth. The sample size (number of units of product) shall be as shown in the table below. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more units of product fail to meet any test requirement specified.

Lot size (yards)	*Sample size*
800 or less	2
801 to 22,000	3
22,001 and up	5

4.4.1 *Test for sulfur dyes.* Sulfur dyes are determined by tests for available sulfur.

4.4.1.1 *Procedure.* Dissolve 100 grams of stannous chloride crystals in 100 ml. of hydrochloric acid (35 percent concentration) and add 50 ml. of distilled water. One to 1.5 grams of the finished cloth, cut into small pieces, shall be used for the test. To preclude interference in dye identification, sample first shall be boiled in distilled water to remove any trace of chemical used in the dyeing process. Place the sample in a test tube of suitable size and cover with the stannous chloride solution. Place a filter paper, which has been moistened with one drop of lead acetate solution, over the mouth of the tube (punch needle hole in paper). Heat the test tube slowly over a low flame until the solution is boiling (prolonged boiling will reduce the sensitivity of the test). A brown stain appearing on the paper indicates the presence of sulfur.

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civilian procurements.)

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2 and 6.2.1).

5.1.1 *Levels A, B, and C.* The cloth, put up in flat folds, shall be packaged in accordance with the applicable requirements for class 3 fabrics of PPP–P–51.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2 and 6.2.1).

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TABLE III. *Tests*

Characteristic	Requirement reference	Test method	Number of determinations	Results reported as
Yarn ply	3.2.1	Visual	1	Pass or fail
Weight in ounces per square yard	Table I	5041		
Yarns per inch:				
Warp	Table I	5050		
Filling	Table I	5050		
Breaking strength, pounds:				
Warp	Table I	5100		
Filling	Table I	5100		
Weave	3.4.1	Visual	1	Pass or fail
Take up warp, percent	3.4.2	4110 [1]	10	Average reported to nearest 0.1 percent.
Warp pattern	3.4.3	Visual	1	Pass or fail
Test for sulfur dyes	3.3	4.4.1	1	Pass or fail
Colorfastness to:				
Light [2]	3.3.2	5660		
Laundering	3.3.2	5610		
Bleaching	3.3.2	5600		
Perspiration [3]	3.3.2	5680 or 5682		
Crocking	3.3.2	5651		
Nonfibrous material content	3.6	2611		
Shrinkage in laundering % [4]	3.7	5550		

[1] The following changes shall be incorporated in the test method:
a. Substitute the word "take-up" wherever the word "'crimp" appears.
b. The test specimen shall be a puckered warp yarn removed from a 14-inch length of cloth.
c. Delete the formula for percent crimp and substitute as follows:

$$\text{Take up, percent} = \frac{L - 10}{L} \times 100$$

L = Total distance between marks under tension.

[2] In the event of a dispute resulting from the test with method 5660 or as a result of suspected anomalous behavior of certain dye types of formulations, the contracting officer may authorize the exposure to natural light in accordance with method 5662.

[3] The transfer cloth shall consist of a warp of polyester fiber or Orlon 81 and a 6-fiber filling, with equal bars of wool, spun nylon, spun rayon, spun silk, spun acetate and bleached cotton. Each 6-fiber repeat shall measure approximately 2 inches and shall be separated by a waste filling stripe.

[4] The three specimens shall be selected from the cloth (unit of product) as follows: One specimen from each side of the cloth within 3 inches of selvage and one specimen from the center of the cloth. No two specimens shall contain the same filling yarns.

5.2.1 *Levels A, B, and C.* The cloth shall be packed in accordance with the applicable requirements of PPP–P–51.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP–P–51.

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for use in clothing items.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type and class of cloth required (see 1.2.1).
(*c*) Color of cloth required (see 3.3).

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(*d*) Color of tight beam warp yarn (see 3.4.3.1 and 3.4.3.2).
(*e*) Color of slack beam warp yarn (see 3.4.3.3).
(*f*) Width of cloth required (see 3.5).
(*g*) Length of piece if other than specified (see 3.8).
(*h*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices, and alternatives, as indicated in PPP–P–51, may be exercised in the preparation for delivery of this cloth.

6.3 **Standard sample.** For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 **Transportation description.** Transportation descriptions, and minimum weights applicable to this commodity are:

Rail:
Cotton, piece goods in the original piece. Carload minimum weight—any quantity.

Motor:
Cotton piece goods, not otherwise indexed. Motor volume minimum weight—any quantity.

6.5 Level B packaging and packing should be used by ordering agencies only when shipment is known to be made to a holding supply depot or warehouse from which partial shipments will be made to ultimate consumers.

NOTICE.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:

Army—M Q
Navy—S MC
Air Force—WADD

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